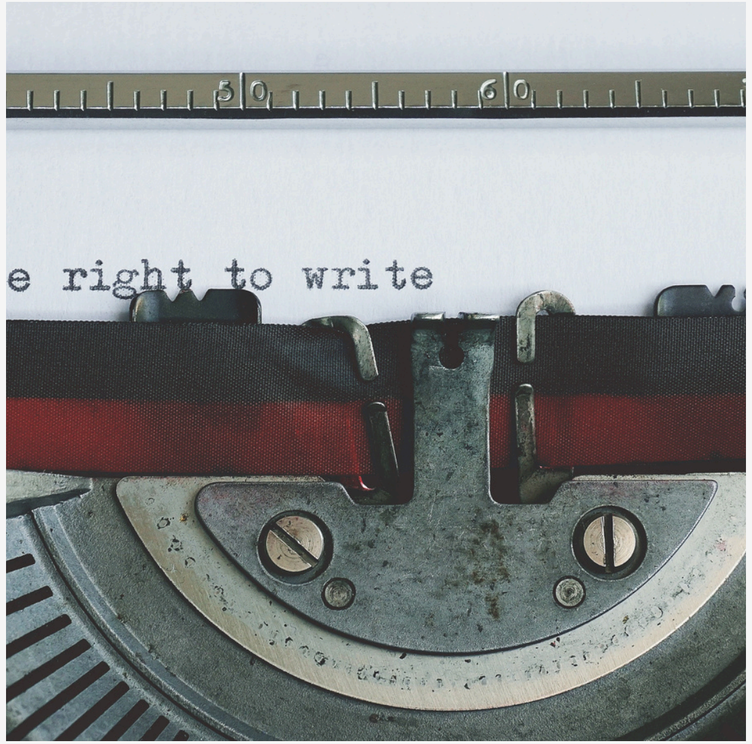
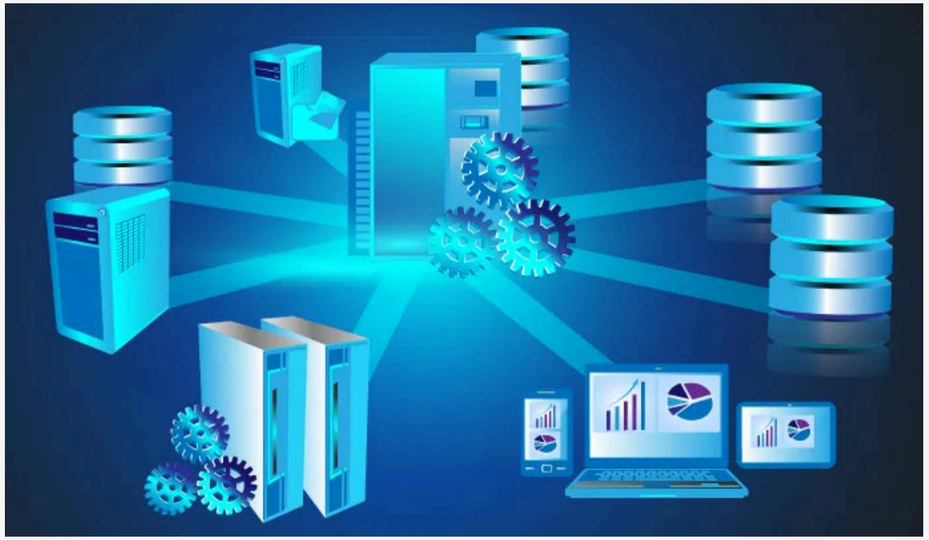
INTRODUCTION TO SQL AND RELATIONAL DATABASES WITH POSTGRESSQL

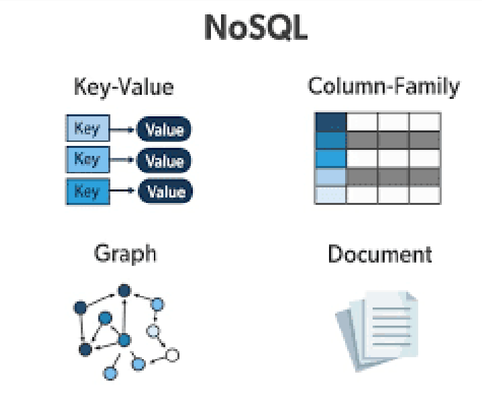
COURSE EXPECTATION

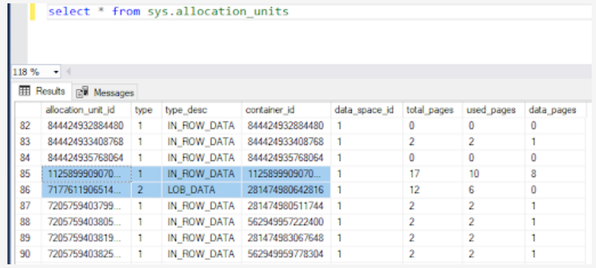


INTRODUCTION TO DATABASE



Database is a collection of data, and are designed to store, manage, and manipulate large volumne of data. Databases come in two main forms known as:





```
select * from sys.allocation_units
```

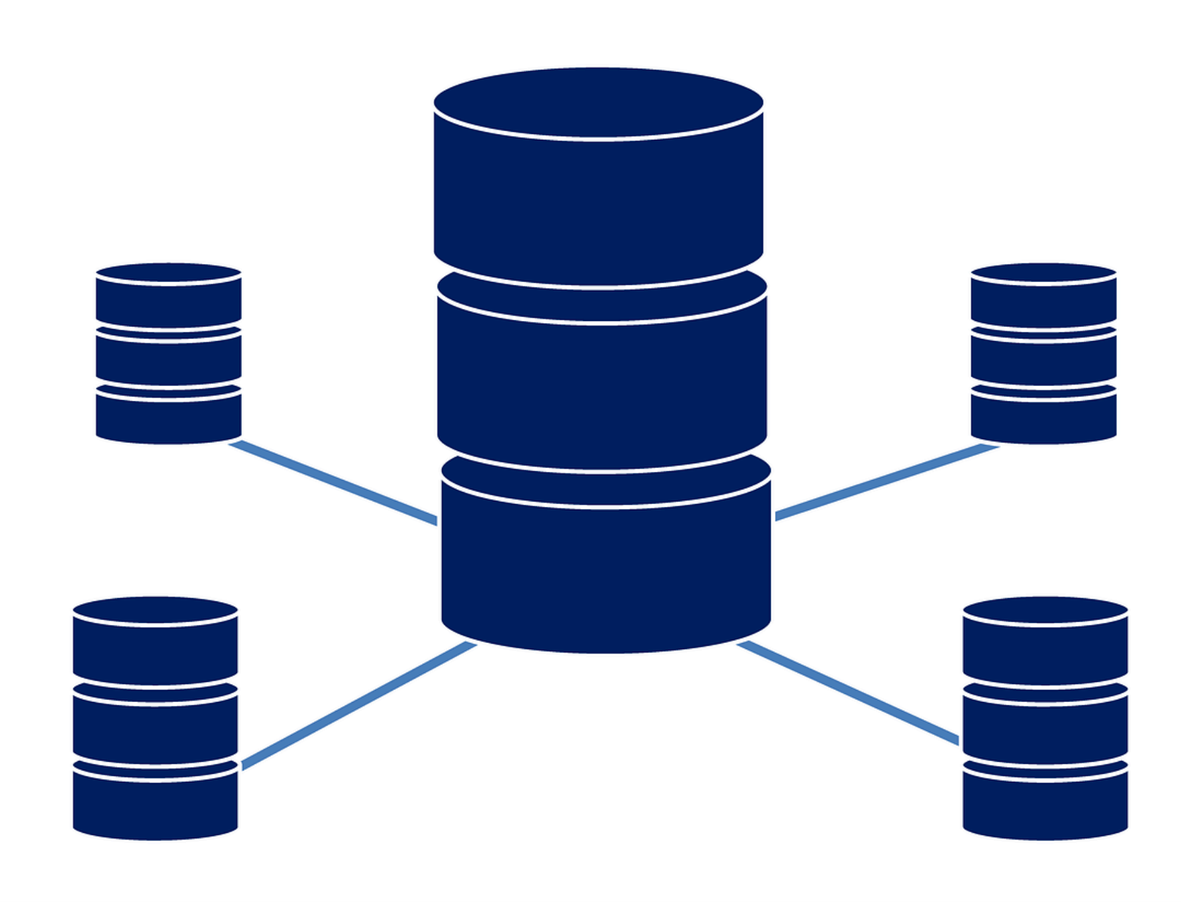
118 %

Results | Messages

	allocation_unit_id	type	type_desc	container_id	data_space_id	total_pages	used_pages	data_pages
82	844424932884480	1	IN_ROW_DATA	844424932884480	1	0	0	0
83	844424933408768	1	IN_ROW_DATA	844424933408768	1	2	2	1
84	844424935768064	1	IN_ROW_DATA	844424935768064	1	0	0	0
85	1125899909070...	1	IN_ROW_DATA	1125899909070...	1	17	10	8
86	7177611906514...	2	LOB_DATA	281474980642816	1	12	6	0
87	7205759403799...	1	IN_ROW_DATA	281474980511744	1	2	2	1
88	7205759403805...	1	IN_ROW_DATA	562949957222400	1	2	2	1
89	7205759403819...	1	IN_ROW_DATA	281474983067648	1	2	2	1
90	7205759403825...	1	IN_ROW_DATA	562949959778304	1	2	2	1

RELATIONAL DATABASES

These are databases that come in the form of tables, like an excel spreadsheet. It is laid out in rows and columns. All relational databases use/ speak a language known as SQL.



Databases and Spreadsheet

Similarities

Tahap Kesadaran

01 Both store data in organized tabular formats — **Data Organization**

02 Both uses rows and columns to structure information — **Informstion structure**

03 Both make room for filtering, sorting snd other basic queries — **Queries**

04 Both can contain multiple sheets/tables — **Number of tables**

Databases and Spreadsheet

Differences

Spreadsheet		Relational Databases
Best when the data observations are in thousands	Scale	Able to handle milllions of observations efficiently
Speed is inversely proportional to amount of data	Performance	Optimized for speed at any scale
Basic file-level protection	Security	High level of access control
No compliance	ACID	Prioritize ACID compliance
Formulas and Functions	Language	SQL
Manual validation and prone to errors	Integrity	Automatic constraints and validation
One or very few concurrent users	Users	Many simultaneous users

The pros of a Database is coherent in its **Structure**

Structure of database

Structure is how data is arranged and organized in database.

Instead of just throwing all data into a single pile, structure ensures that data is organized into:

1. Tables
2. Columns
3. Rows
4. Rules

Why Structure Matter

Integrity: Structure products data quality through certain standard rules:

Example:

A student's age cannot be negative

Duplicated email must be prevented

Scalability: Structured database takes into account the possibility of growing from hundreds to trillions of records without degrading in performance.

Consistency: Structure ensures that the data is entered the same way every.

Eg. Phone numbers and dates

233-1234567

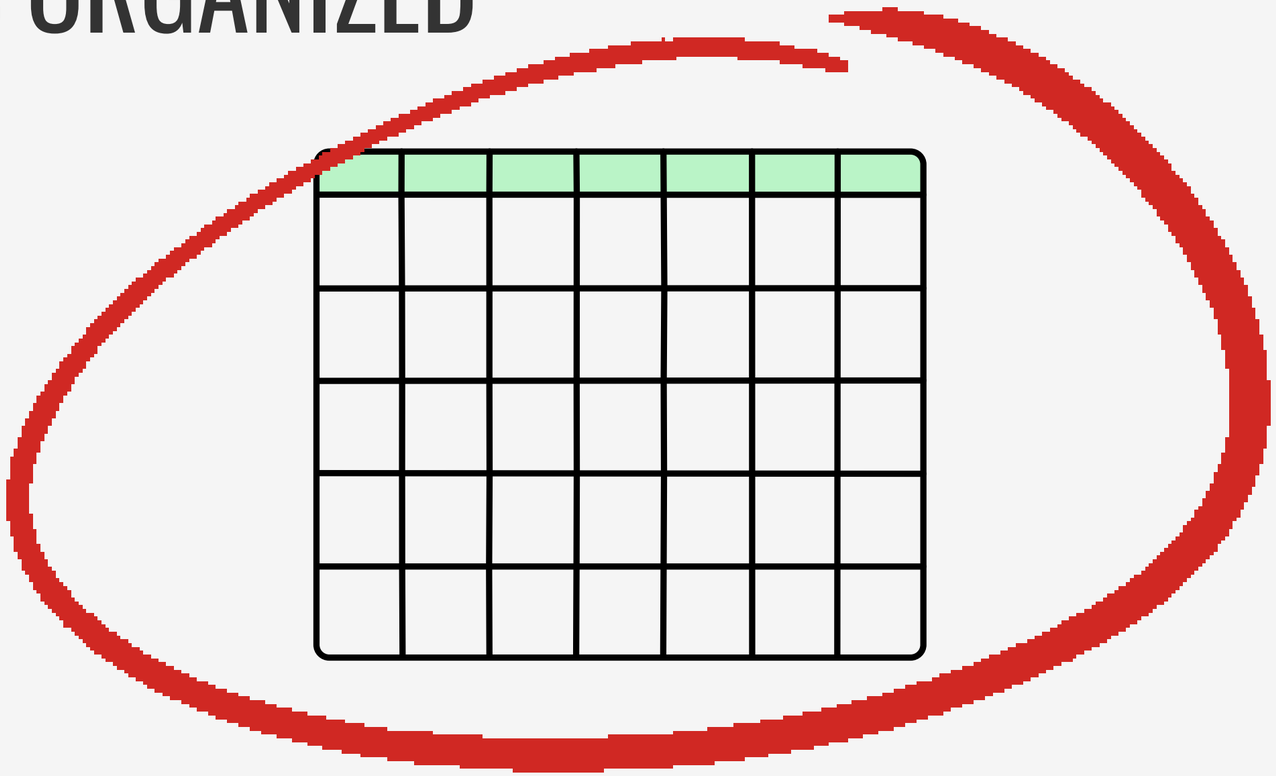
DD-MM-YYYY

HOW DATA IS ORGANIZED

Tables as Entities

Rows as Records

Columns as Attributes



Relational model organizes data into interconnected tables where each table represents entities. So, instead of storing customer and product information in every order row:

Customers once in **‘Customer Table’**
Product once in **‘Product Table’**
Then the **‘Order Table’** just reference them

KEYS IN DATABASE

Special Columns (combination of columns) used to identify rows and link tables together

Natural
Existing real-world data that is naturally unique

Surrogate
Artificially created primary keys with no real-world meaning

Primary Key

Foreign Key

One-to-One
One row in the parent relates to one row in the child table

One-to-Many
One row in the parent relates to one row in the child table

Many-to-Many
Many rows in the parent relates to many rows in the child table

Primary Key	Main identifyer. No two rows can have same key value
Foreign Key	Column(s) that reference the primary key of another table
Candidate Key	Any column (set of columns) that can act as primary key
Composite Key	Primary key made of multiple columns together
Super Key	Any set of columns including the primary key that can uniquely identify a row

NORMALIZATION IN DATABASE

The process of organizing data to reduce redundancy and improve consistency

STUDENT_ID	STUDENT_NAME	STUDENT_EMAIL	COURSE_ID	COURSE_NAME	INSTRUCTOR	GRADE
1	Grace Koku	gk@gmail.com	MATH 210	Calculus	Larry Klutse	84
2	Grace Koku	gk@gmail.com	MATH 212	Linear Algebra	Bismarck Duah	91
3	Grace Koku	gk@gmail.com	MATH 214	Abstract Algebra	Juliet Ansah	75
4	Bright Andor	ba@gmail.com	MATH 212	Linear Algebra	Bismarck Duah	88
5	Bright Andor	ba@gmail.com	MATH 210	Calculus	Larry Klutse	90

Wasting storage due to repeaded information (redundancy)

Future changes lead to many updates (update anomalies)

Inability to introduce new item due to incomplete data (insert anomalies)

Unintended data loss (delete anomalies)

STRUCTURE OF DATABASE

Structure is how data is arranged and organized in database.

Instead of just throwing all data into a single pile, structure ensures that data is organized into:

1. Tables
2. Columns
3. Rows
4. Rules

Consistency

This means uniform data entry. For instance, ensuring that all dates are in the same format

Integrity

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis vel fringilla nisl. Maecenas pellentesque dui at eros tristique, a placerat ligula pretium.

Scalability

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INTRODUCTION TO POSTGRESQL

PostgreSQL